

Final Project Proposal

About The Data Collection

For this project, I will collect daily data over a two-week period (14 days) on my personal budgeting and spending habits. Each day, I will record the amount I plan to spend (daily budget) and the amount I actually spend (daily spending). The dataset is quantitative and consists of two numerical values per day (in USD).

Variables in The Data Collection

- **Independent Variable:** Daily Budget – This represents the amount I plan to spend each day, determined in advance.
- **Dependent Variable:** Daily Spending – This is the actual amount spent each day, which may be influenced by various factors such as impulse purchases or unforeseen expenses.

Research Question

Is my daily budget significantly greater than my daily spending?

- This question is meaningful because it examines whether my financial planning is conservative or if I tend to spend close to (or even exceed) my planned amounts.

Understanding this relationship is important for improving personal financial management.

Justification for Data and Question Selection

I chose this dataset because personal budgeting is a critical component of financial planning and discipline a topic highly relevant to my studies in Finance and Business Analytics. Additionally, this project will help me understand real-life spending behaviors while applying statistical

analysis techniques. The insights gained can also be valuable for future roles in financial planning or business analytics.

Graphical Representations

I plan to use the following visualizations:

- **Line Graphs:** To display trends in both daily budget and actual spending over the two-week period.
- **Bar Charts:** To compare the daily values side by side.
- **Box Plots:** To visualize the distribution of planned and actual spending and highlight any outliers.

These visualizations are appropriate because they provide a clear, intuitive view of the data trends and differences between planned and actual spending.

Hypothesis Testing

I will conduct a paired t-test to compare the two sets of data. The hypotheses for this test are:

- **Null Hypothesis (H_0):** The mean daily budget is equal to the mean daily spending.

$$H_0: \mu_{\text{budget}} = \mu_{\text{spending}}$$

- **Alternative Hypothesis (H_1):** The mean daily budget is greater than the mean daily spending.

$$H_1: \mu_{\text{budget}} > \mu_{\text{spending}}$$

By testing these hypotheses, I aim to determine whether my budgeting practice is conservative enough to account for potential overspending, which is an important aspect of financial discipline.